

Exploratory Project Report

submitted in partial fulfillment for the award of
Integrated Dual Degree (B.Tech + M.Tech)

ML/AI Based Modelling for Additive Manufacturing Process Optimisation

*Physics-Informed Bayesian Machine Learning for Residual Stress
Prediction in LPBF*

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Certificate

This is to certify that the project titled “**ML/AI Based Modelling for Additive Manufacturing Process Optimisation**”, submitted by the following students, to the Indian Institute of Technology (Banaras Hindu University), Varanasi, in partial fulfillment for the award of the Integrated Dual Degree (B.Tech + M.Tech), is a bona fide record of work done by them under my supervision.

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Declaration

We, the undersigned, certify that the work embodied in this project is our own bona fide work carried out under the supervision of Dr. Pawan Sharma. The matter embodied in this report has not been submitted for the award of any other degree or diploma. We declare that we have faithfully acknowledged and cited all researchers whose works have been utilised in this report.

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Abstract

Laser Powder Bed Fusion (LPBF) is a powerful additive manufacturing (3D printing) technique that can create complex metal parts layer-by-layer using a laser. One of its biggest challenges is *residual stress* — internal forces trapped inside a part after it cools, which can cause warping, cracking, and failure.

This project builds a **physics-informed machine learning pipeline** to predict residual stress in MS300 Maraging Steel parts as a function of three laser process parameters: power, scan speed, and hatch spacing. Rather than using a purely data-driven (black-box) approach, we embed an established thermal physics model (the Eagar-Tsai model) inside a probabilistic machine learning algorithm (Gaussian Process Regression), giving accurate predictions together with uncertainty estimates.

Ground-truth stress data was generated using **Simufact Additive**, a commercial finite element simulation tool. A three-layer cascaded machine learning model (called M5) was trained on this data. A global optimisation algorithm (Differential Evolution) then searches the parameter space to find laser settings that minimise residual stress while avoiding harmful process regimes.

The pipeline was deployed as a graphical web application, allowing users to input parameters and receive predicted stress values, confidence intervals, and optimised parameter recommendations.

Keywords: LPBF, Additive Manufacturing, Residual Stress, Gaussian Process Regression, Eagar-Tsai Model, Bayesian Machine Learning, Simufact, Differential Evolution, MS300 Maraging Steel.

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Chapter 1

Introduction

1.1 Background and Motivation

Modern engineering demands components with complex internal geometries — think of turbine blades with internal cooling channels, or lightweight lattice structures for aerospace. Traditional manufacturing (cutting, forging, casting) often cannot produce these shapes economically. **Additive Manufacturing (AM)**, commonly called 3D printing for metals, solves this problem by building parts directly from a digital model, one thin layer at a time.

Among all metal AM processes, **Laser Powder Bed Fusion (LPBF)** has become one of the most widely used industrial methods. It can process high-strength alloys including titanium, stainless steel, aluminium, and maraging steels. However, LPBF comes with a critical challenge: because the laser heats and cools tiny regions of metal extremely rapidly, large internal stresses build up inside the finished part. These are called **residual stresses**.

Why do residual stresses matter?

- They can cause parts to **warp** and lift off the build platform during printing.
- They reduce **fatigue life** — parts fail sooner under repeated loads.
- They can cause **cracking** during or after the build.
- They often require expensive post-processing (heat treatment) to relieve.

Predicting and controlling residual stress is therefore extremely important for producing reliable LPBF parts.

1.2 Problem Statement

The laser process parameters (power, speed, hatch spacing) can be adjusted by the machine operator, but there is no simple equation connecting these settings to the resulting residual stress. Traditional methods rely on either:

1. **Trial and error:** printing many samples and measuring them, which is expensive and slow.
2. **Full finite element simulation:** accurate but can take hours per simulation run.

Neither approach can quickly guide an operator towards optimal settings. Our project addresses this gap.

1.3 Project Objectives

1. Understand the LPBF process and how process parameters affect residual stress.
2. Generate a training dataset of residual stress values using Simufact Additive simulation.
3. Build a physics-informed machine learning model that can *rapidly* predict residual stress for any given set of process parameters.
4. Develop an optimisation algorithm to automatically find process parameters that minimise residual stress.
5. Deploy the system as a user-friendly graphical interface (GUI).

1.4 Material: MS300 Maraging Steel

We chose **MS300 Maraging Steel** for this study. This is a high-strength, low-carbon steel widely used in aerospace, tooling, and defence applications. Its key thermal properties relevant to our model are listed in Table 1.1.

Table 1.1: MS300 Maraging Steel — Thermal Properties Used in the Model

Property	Symbol	Value	Unit
Thermal conductivity	k	25.0	W/(m·K)
Density	ρ	8000	kg/m ³
Specific heat capacity	c_p	460	J/(kg·K)
Liquidus temperature	T_L	1415	°C
Laser absorptivity	η	0.40	—
Laser beam radius	r	35	μm

Chapter 2

How Laser Powder Bed Fusion Works

This chapter explains the LPBF process in detail, step by step. Understanding the physical process is essential before we can model it mathematically.

2.1 The LPBF Machine

An LPBF machine has the following key components:

- **Laser source:** Typically a fibre laser (200–400 W) focused to a spot of 50–100 μm diameter.
- **Galvo mirrors:** Rapidly steer the laser beam across the build area.
- **Powder delivery system:** A hopper and recoater blade that spread fresh powder.
- **Build platform:** A movable plate that descends after each layer.
- **Process chamber:** Sealed and filled with inert gas (argon or nitrogen) to prevent oxidation.

2.2 Step-by-Step Build Process

The LPBF Cycle (repeated for every layer):

1. The **recoater blade** spreads a thin, uniform layer of metal powder (typically 20–60 μm thick) across the build platform.
2. The **laser beam** scans across the cross-section of the part as defined by the digital model (CAD file sliced into layers).
3. Wherever the laser hits, the powder **melts and fuses**. A tiny pool of liquid metal (the melt pool) forms and solidifies within milliseconds as the laser moves on.

4. After scanning is complete, the build platform **descends** by exactly one layer thickness.
5. Steps 1–4 repeat until the entire part is complete.

2.3 Key Process Parameters

The quality, strength, and stress state of the final part depend heavily on four parameters that the operator sets before printing:

Table 2.1: LPBF Process Parameters and Their Roles

Parameter	Symbol	Typical Range	Effect
Laser Power	P (W)	100–400 W	More power $\rightarrow$ more energy into powder, deeper melt
Scan Speed	v (mm/s)	200–1200 mm/s	Faster speed $\rightarrow$ less time over each spot, shallower melt
Hatch Spacing	h (mm)	0.05–0.15 mm	Gap between scan lines; affects overlap and energy distribution
Layer Thickness	t (mm)	0.02–0.06 mm	Thicker layers are faster but risk incomplete fusion (fixed at 0.04 mm in our study)

2.4 How Residual Stress is Generated

Residual stress in LPBF is not caused by any single event but is the result of three mechanisms acting together:

2.4.1 Thermal Gradient Mechanism (TGM)

This is the dominant mechanism. When the laser hits the powder, it heats a tiny spot to temperatures above 1415°C (the melting point). The surrounding material is still cold. The hot material tries to expand, but the cold surrounding material prevents this — so the hot zone is put under *compression*.

When the laser moves away, that spot cools very rapidly. The material that was compressed now tries to contract, but again is constrained. The result is tensile (pulling) residual stress. This is the main reason why LPBF parts tend to have tensile residual stresses near the surface.

2.4.2 Repeated Heating and Cooling

Every new layer that is melted also reheats the layer below it slightly. This repeated thermal cycling accumulates and modifies the stress state from earlier layers. The net stress field is therefore a complex superposition of many heating/cooling events.

2.4.3 Scan Pattern Effects

The direction in which the laser scans (horizontal, vertical, rotated 67° between layers) also affects the direction and magnitude of stress. Longer scan lines produce higher stresses along the scan direction.

2.5 Process Defect Regimes

Not all combinations of laser power and speed produce good parts. There are three regimes, distinguished by a quantity called the Volumetric Energy Density (VED):

$$\text{VED} = \frac{P}{v \cdot h \cdot t} \quad [\text{J/mm}^3] \quad (2.1)$$

Table 2.2: LPBF Process Quality Regimes

Regime	VED Range	Description
Lack of Fusion	Too low (<50 J/mm ³)	Not enough energy to fully melt the powder. Layers don't bond properly. Porous, weak parts.
Good Processing Window	50–120 J/mm ³	Stable melt pool, good density, acceptable stress. Target zone for most alloys.
Keyholing	Too high (>120 J/mm ³)	Too much energy causes deep vapour cavities. Gas gets trapped, forming round pores.

Our optimisation algorithm explicitly avoids the keyholing regime when recommending process parameters.

Chapter 3

Generating Training Data: Simufact Additive

3.1 Why We Need Simulation

Our machine learning model needs *training data* — pairs of (process parameters, residual stress) from which it can learn. Physical experiments (printing real samples and measuring stress using X-ray diffraction) are accurate but extremely time-consuming and costly. A single parameter combination could take several hours to print and measure.

Simufact Additive is a commercial finite element analysis (FEA) software specifically designed for additive manufacturing. It allows us to simulate the LPBF process and extract residual stress values in a matter of minutes per simulation run. This makes it practical to generate a large dataset covering many combinations of power, speed, and hatch spacing.

3.2 What is Finite Element Analysis?

FEA is a numerical method that divides a complex object into thousands of small elements (like a 3D mesh). For each element, the software solves the equations of heat transfer and mechanics. The solution gives temperature and stress at every point in the part.

Think of it like this: instead of solving one impossible equation for the whole part, you solve many simpler equations for thousands of tiny pieces, then stitch the solutions together.

3.3 How Simufact Models the LPBF Process

Simufact Additive performs the following steps for each simulation:

1. **Layer-by-layer thermal simulation:** The laser is modelled as a moving heat

source. For each layer, the software computes the temperature at every point as the laser scans and the material heats and cools.

2. **Thermo-mechanical coupling:** Temperature changes cause the material to expand and contract. The software computes the resulting mechanical stresses and deformations, using temperature-dependent material properties (the yield strength and stiffness of MS300 change with temperature).
3. **Stress extraction:** Once the full build is simulated and cooled to room temperature, the residual stress at a specific depth ($z = 3.861$ mm from the base plate) is extracted. This depth was chosen to represent the mid-section of a typical test specimen.
4. **Parameter sweep:** The above is repeated for different combinations of laser power, scan speed, and hatch spacing to build our training dataset.

3.4 Dataset Description

The final dataset contains columns as described in Table 3.1.

Table 3.1: Description of the Training Dataset

Column Name	Description
Power_W	Laser power (W)
Speed_mms	Scan speed (mm/s)
Hatch_mm	Hatch spacing (mm)
ED_Jmm3	Volumetric Energy Density (computed)
Res_Stress	Normalised residual stress at $z = 3.861$ mm
VED_Prior_Stress	Physics-based sigmoid prior prediction
Stress_Residual	Difference: actual – prior

Chapter 4

The Eagar-Tsai Thermal Model

4.1 Why We Need a Thermal Model

Before we can do machine learning, we want to extract physically meaningful *features* from the process parameters. Rather than feeding just (Power, Speed, Hatch) into our ML model, we also compute the shape of the melt pool — how wide and deep the melted region is. This gives the model extra information that helps it understand the physics.

The **Eagar-Tsai model** provides an analytical (formula-based) way to compute the temperature field around a moving laser. From this temperature field, we can determine where the metal melts and therefore estimate the melt pool geometry.

4.2 Physical Basis

The Eagar-Tsai model is an extension of an older model called Rosenthal’s solution. Rosenthal modelled the laser as a *point* heat source; Eagar and Tsai improved this by modelling the laser as a *Gaussian* (bell-shaped) heat source, which is physically more accurate.

The model assumes:

- The material extends infinitely in all directions below the surface (semi-infinite solid).
- Heat travels by conduction only (radiation and convection within the melt pool are ignored).
- The laser moves at constant speed v in the x -direction.
- Material properties (k , ρ , c_p) are constant (not temperature-dependent).

4.3 Temperature Field

We work in a coordinate frame that moves with the laser. Define $\xi = x - vt$ as the distance behind the laser. The temperature rise ΔT at any point (ξ, y, z) is given by the Eagar-Tsai integral:

$$\Delta T(\xi, y, z) = \frac{\sqrt{2} \eta P \alpha}{\pi^{3/2} k} \int_0^\infty \frac{\exp\left(-\frac{\xi^2}{2d_{xy}(\tau)} - \frac{y^2}{2d_{xy}(\tau)} - \frac{z^2}{d_z(\tau)} - \frac{v\xi}{2\alpha} - \frac{v^2\tau}{4\alpha}\right)}{d_{xy}(\tau) \sqrt{2\alpha\tau}} d\tau \quad (4.1)$$

where the time-dependent diffusion widths are:

$$d_{xy}(\tau) = 2\alpha\tau + r^2 \quad (\text{lateral: combines thermal diffusion and beam radius}) \quad (4.2)$$

$$d_z(\tau) = 4\alpha\tau \quad (\text{axial: thermal diffusion only}) \quad (4.3)$$

and the thermal diffusivity is:

$$\alpha = \frac{k}{\rho c_p} = \frac{25.0}{8000 \times 460} \approx 6.79 \times 10^{-6} \text{ m}^2/\text{s} \quad (4.4)$$

Plain-language explanation of Equation 4.1:

The integral sums up the contribution of the laser heat source at all past times τ . The exponential terms describe how heat diffuses outward from the heat source: the further a point is from the laser (in ξ , y , or z), the smaller the temperature rise. The term $d_{xy}(\tau)$ grows with time because heat spreads laterally as the laser moves forward.

4.4 Finding the Melt Pool

The melt pool boundary is the set of all points where ΔT equals exactly the melting temperature rise:

$$\Delta T_{\text{melt}} = T_L - T_0 = 1415 - 80 = 1335^\circ\text{C} \quad (4.5)$$

This isotherm (constant temperature surface) defines the shape of the molten region. The numerical integration is performed using standard scientific computing tools over a time range of $\tau \in [10^{-9}, 5 \times 10^{-3}]$ seconds.

Chapter 5

Physics Prior: VED-Based Stress Model

5.1 The Key Idea

Machine learning models work best when they have a good *starting point* — some knowledge of the underlying trend before they see any data. We provide this using a physics-based prior model.

We observed that **Volumetric Energy Density** (VED, Equation 2.1) captures the dominant influence of process parameters on residual stress. Higher VED generally means more heat input, higher peak temperatures, larger thermal gradients, and thus higher residual stress. This trend can be modelled with a **sigmoid** (S-shaped) function.

5.2 Sigmoid Prior Models

We fit three different sigmoid functions to the VED vs. stress data and select the best one.

5.2.1 Simple Sigmoid

The most basic model: stress increases smoothly from zero to a maximum as VED increases.

$$\sigma_{\text{prior}} = \frac{S_{\text{max}}}{1 + e^{-k_r(\text{VED} - \text{VED}_{1/2})}} \quad (5.1)$$

Parameters: S_{max} (maximum stress), k_r (steepness of rise), $\text{VED}_{1/2}$ (VED at half-maximum stress).

5.2.2 Bell Sigmoid

At very high VED (keyholing regime), the extreme heat can partially relieve stress through annealing. This model adds a decreasing quadratic term:

$$\sigma_{\text{prior}} = \frac{S_{\text{max}}}{1 + e^{-k_r(\text{VED} - \text{VED}_{\text{opt}})}} - k_f \cdot \max(\text{VED} - \text{VED}_{\text{opt}}, 0)^2 \quad (5.2)$$

5.2.3 Double Sigmoid

Models two distinct physical regimes (under-melted and keyholing) separately:

$$\sigma_{\text{prior}} = \frac{S_{\infty}}{1 + e^{-k_1(\text{VED} - \text{VED}_c)}} - \frac{\Delta}{1 + e^{-k_2(\text{VED} - \text{VED}_{\text{key}})}} \quad (5.3)$$

The second term captures stress reduction in the keyholing regime.

5.3 Residual Signal for ML

Once the best sigmoid is fitted, we compute the *residual* signal — the difference between the actual measured/simulated stress and what the physics prior predicts:

$$\sigma_{\text{residual}} = \sigma_{\text{measured}} - \sigma_{\text{prior}}(\text{VED}) \quad (5.4)$$

The machine learning model learns only this residual. This is the heart of our physics-informed design. Because the prior already captures the dominant VED trend, the residual is a much smaller, smoother signal that the ML model can learn efficiently, even from a limited dataset.

Chapter 6

The Machine Learning Model: Gaussian Process Regression

6.1 Why Gaussian Process Regression?

We chose **Gaussian Process Regression (GPR)** as our machine learning algorithm for several reasons:

- **Small dataset:** We have a limited number of simulation runs. GPR works well with small datasets; neural networks typically need much more data.
- **Uncertainty quantification:** GPR does not just predict a single number — it gives a full probability distribution over possible values. This means we know *how confident* the model is in each prediction.
- **Interpretability:** The kernel function explicitly encodes assumptions about how smooth and how nonlinear the relationship is.

6.2 What is a Gaussian Process?

A **Gaussian Process (GP)** is a way to define a probability distribution over *functions*. Instead of saying “the stress is exactly 250 MPa”, a GP says “the stress is probably somewhere between 230 and 270 MPa, with 250 being most likely”.

Formally: given training data $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^n$ where $y_i = f(\mathbf{x}_i) + \varepsilon$ and $\varepsilon \sim \mathcal{N}(0, \sigma_n^2)$ is measurement noise, the **posterior predictive distribution** at a new point $\mathbf{x}^*$ is:

$$p(f(\mathbf{x}^*) \mid \mathbf{x}^*, \mathcal{D}) = \mathcal{N}(\mu(\mathbf{x}^*), \sigma^2(\mathbf{x}^*)) \quad (6.1)$$

The predictive **mean** (our best guess) and **variance** (our uncertainty) are:

$$\mu(\mathbf{x}^*) = \mathbf{k}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} \quad (6.2)$$

$$\sigma^2(\mathbf{x}^*) = k(\mathbf{x}^*, \mathbf{x}^*) - \mathbf{k}_*^T (\mathbf{K} + \sigma_n^2 \mathbf{I})^{-1} \mathbf{k}_* \quad (6.3)$$

where $\mathbf{K}$ is the $n \times n$ kernel matrix with $K_{ij} = k(\mathbf{x}_i, \mathbf{x}_j)$, and $\mathbf{k}_* = [k(\mathbf{x}^*, \mathbf{x}_1), \dots, k(\mathbf{x}^*, \mathbf{x}_n)]^T$.

Plain-language meaning:

Equation 6.2: Our prediction is a weighted average of all training observations. Training points with high $k(\mathbf{x}^*, \mathbf{x}_i)$ (i.e., similar to $\mathbf{x}^*$) get more weight.

Equation 6.3: Our uncertainty equals the prior uncertainty *minus* the information we gained from training data. Far from all training points, variance is high; near training points, variance is low.

6.3 The Kernel Function

The **kernel function** $k(\mathbf{x}, \mathbf{x}')$ measures how similar two input points are. It encodes our assumptions about the function we are trying to learn. We use an additive combination of three kernels:

$$k(\mathbf{x}, \mathbf{x}') = k_{\text{RBF}}(\mathbf{x}, \mathbf{x}') + k_{\text{DP}}(\mathbf{x}, \mathbf{x}') + k_{\text{W}}(\mathbf{x}, \mathbf{x}') \quad (6.4)$$

$$k_{\text{RBF}}(\mathbf{x}, \mathbf{x}') = \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}'\|^2}{2\ell^2}\right), \quad k_{\text{DP}}(\mathbf{x}, \mathbf{x}') = \sigma_0^2 + \mathbf{x} \cdot \mathbf{x}', \quad k_{\text{W}}(\mathbf{x}, \mathbf{x}') = \sigma_n^2 \delta(\mathbf{x}, \mathbf{x}') \quad (6.5)$$

Table 6.1: Kernel Components and Their Physical Meaning

Kernel	Name	Physical Role
k_{RBF}	Radial Basis Function	Captures smooth, nonlinear effects. The length-scale ℓ controls how quickly the function varies.
k_{DP}	Dot Product	Captures global linear trends, e.g. stress increasing with power.
k_{W}	White Noise	Represents noise in the data and unmodelled physics. Prevents overfitting.

6.4 Hyperparameter Optimisation

The kernel has unknown parameters (length-scale ℓ , σ_0 , σ_n). These are found automatically by maximising the **log marginal likelihood**:

$$\log p(\mathbf{y} \mid \mathbf{X}, \theta) = -\frac{1}{2} \mathbf{y}^T (\mathbf{K}_\theta + \sigma_n^2 \mathbf{I})^{-1} \mathbf{y} - \frac{1}{2} \log |\mathbf{K}_\theta + \sigma_n^2 \mathbf{I}| - \frac{n}{2} \log 2\pi \quad (6.6)$$

This is called *type-II maximum likelihood*. It balances data fit (first term) against model complexity (second term), automatically preventing overfitting. The optimisation is run 5 times from different starting points to avoid getting stuck in local optima.

6.5 Confidence Intervals

The 95% confidence interval for any new prediction is:

$$\text{CI}_{95\%} = [\mu(\mathbf{x}^*) - 1.96 \sigma(\mathbf{x}^*), \mu(\mathbf{x}^*) + 1.96 \sigma(\mathbf{x}^*)] \quad (6.7)$$

This tells the user: “we are 95% confident the true stress lies within this range.”

Chapter 7

The M5 Three-Layer Prediction Architecture

7.1 Why Three Layers?

A single GPR trained on (Power, Speed, Hatch) $\rightarrow$ Stress misses an important subtlety: two parameter sets can have the *same* VED but produce very different residual stresses because their melt pool geometries differ. For example, high power at high speed can produce the same VED as low power at low speed, but the first creates a wide, shallow melt pool while the second creates a narrow, deep one. These have completely different thermal histories and stress states.

By explicitly computing the melt pool dimensions as intermediate features and passing them through the prediction cascade, the M5 model gives the final stress predictor the geometric information it needs.

7.2 The M5 Cascade

The three layers predict in sequence:

$$\underbrace{(P, v, h)}_{\text{inputs}} \xrightarrow{\text{ET model}} (W_{ET}, D_{ET}) \xrightarrow{\text{GPR 1}} W_{\text{corr}} \xrightarrow{\text{GPR 2}} D_{\text{corr}} \xrightarrow{\text{GPR 3}} \sigma_{\text{disc}} \xrightarrow{+ \sigma_{\text{prior}}} \hat{\sigma} \quad (7.1)$$

- **Layer 1 (GPR₁):** Takes (Power, Speed, Hatch) as input and predicts the corrected melt pool width W_{corr} .
- **Layer 2 (GPR₂):** Takes (Power, Speed, Hatch, W_{corr}) and predicts the corrected melt pool depth D_{corr} . Width is included because keyholing is a coupled phenomenon.

- **Layer 3 (GPR₃):** Takes (Power, Speed, Hatch, W_{corr} , D_{corr}) and predicts the stress residual σ_{disc} from Equation 5.4.

7.3 Final Prediction

The final predicted stress adds the physics prior back to the ML correction:

$$\hat{\sigma} = \underbrace{\text{GPR}_3(P, v, h, W_{\text{corr}}, D_{\text{corr}})}_{\text{ML residual}} + \underbrace{\sigma_{\text{prior}}\left(\frac{P}{v \cdot h \cdot t}\right)}_{\text{Physics prior}} \quad (7.2)$$

Chapter 8

Process Parameter Optimisation

8.1 The Optimisation Problem

Given our trained M5 model, we want to find the combination of (Power, Speed, Hatch) that produces a target residual stress level — ideally as low as possible — while staying within a safe process window (avoiding keyholing).

Mathematically:

$$(P^*, v^*, h^*) = \arg \min_{(P, v, h) \in \mathcal{B}} [|\hat{\sigma}(P, v, h) - \sigma_{\text{target}}| + w_{kh} \cdot \mathbf{1}[W/D < 1.5] + w_{\text{unc}} \cdot \hat{\sigma}_{\text{std}}] \quad (8.1)$$

where $\mathcal{B}$ is the feasible parameter range, the keyhole penalty term prevents recommending parameters that lead to defective parts, and the uncertainty term encourages solutions in well-explored regions.

8.2 Differential Evolution

We use **Differential Evolution (DE)**, a type of evolutionary algorithm that mimics natural selection to search a large parameter space efficiently. It does not require gradients, so it can handle non-smooth objective functions and is much less likely to get stuck in local minima than gradient-based methods.

How DE works (simplified):

1. Start with a *population* of 15 random parameter combinations, spread across the full feasible range.
2. For each candidate, create a *mutant* by combining it mathematically with two other candidates from the population.

3. Test whether the mutant is better (has a lower cost). If yes, replace the original candidate.
4. Repeat for up to 200 generations.
5. Polish the best result using a standard gradient-based local search.

8.3 Adaptive Cost Function

The cost function adapts based on how close the current best solution is to the target:

$$\text{cost}(P, v, h) = |\hat{\sigma} - \sigma_{\text{target}}| + w_{kh} \cdot |\hat{\sigma}| \cdot \mathbf{1}[W/D < 1.5] + w_{\text{unc}} \cdot \hat{\sigma}_{\text{std}} \quad (8.2)$$

Table 8.1: Adaptive Weights in the Cost Function

Phase	Condition	w_{kh}	w_{unc}
Sprint	Gap > 15% of full stress range	0.8	0.02
Fine	Gap $\leq$ 15% of full stress range	0.2	0.15

In the Sprint phase, the optimiser focuses on getting close to the target quickly, applying a strong keyhole penalty. In the Fine phase, it focuses on reducing uncertainty, which is more important for fine-tuning.

After global optimisation, a local neighbourhood scan of 600 nearby points produces a ranked list of the top 10 candidate parameter sets. Each is flagged if it falls in the keyhole zone or has high uncertainty.

Chapter 9

Model Evaluation Metrics

We evaluate model accuracy using four complementary metrics:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (9.1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (9.2)$$

$$r_s = 1 - \frac{6 \sum d_i^2}{n(n^2 - 1)} \quad (\text{Spearman rank correlation}) \quad (9.3)$$

$$\tau = \frac{(\text{concordant pairs}) - (\text{discordant pairs})}{\binom{n}{2}} \quad (\text{Kendall's tau}) \quad (9.4)$$

Table 9.1: Explanation of Evaluation Metrics

Metric	What it measures
MAE	Average error in the same units as stress. Easy to interpret.
RMSE	Like MAE but penalises large errors more heavily. Sensitive to outliers.
Spearman r_s	Whether the model correctly <i>ranks</i> different parameter sets by stress level, even if the absolute values are slightly off. Robust to outliers.
Kendall τ	Similar to Spearman; measures the fraction of correctly ordered pairs.

We prefer rank-based metrics (r_s , τ) because residual stress data from near-keyholing conditions can contain extreme outlier values that distort Pearson correlation.

Chapter 10

Graphical Interface and Key Discoveries

10.1 The Graphical User Interface (GUI)

We developed a graphical user interface (GUI) so that users — including machine operators with no machine learning background — can use our pipeline easily.

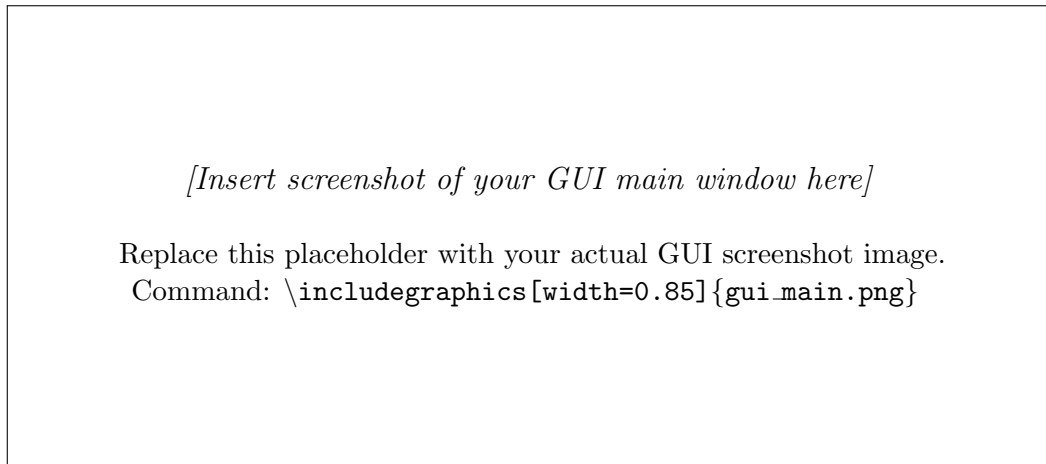


Figure 10.1: Main window of the ML/AI residual stress prediction application

10.1.1 Input Panel

The user enters three process parameters:

- **Laser Power** (W) — slider or numeric field
- **Scan Speed** (mm/s) — slider or numeric field
- **Hatch Spacing** (mm) — slider or numeric field

10.1.2 Output Panel

After clicking “Predict”, the application displays:

- **Predicted residual stress** (normalised value)
- **95% confidence interval** (upper and lower bounds)
- **Process quality flag:** Safe / Keyhole risk / Lack-of-fusion risk
- **VED value** and which process regime it falls in

10.1.3 Optimisation Tab

The user can specify a target stress level and click “Optimise”. The application runs Differential Evolution and returns the top 10 recommended parameter sets, ranked by expected stress with uncertainty flags.

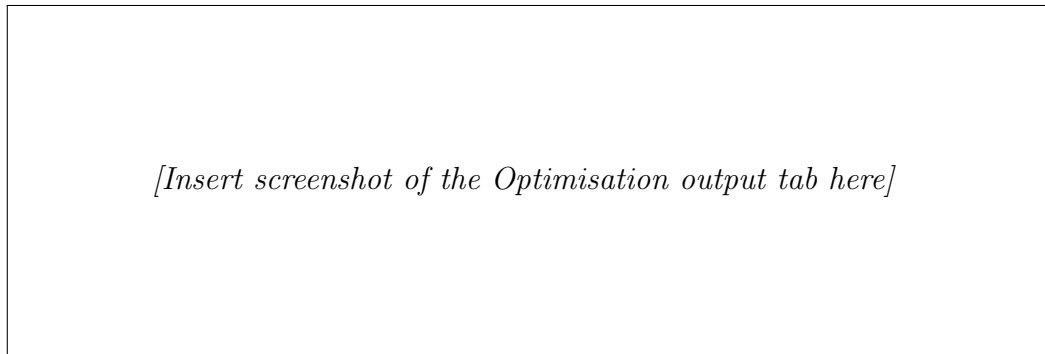


Figure 10.2: Optimisation output panel showing top-10 recommended parameter sets

10.2 Key Graphical Discoveries

10.2.1 VED vs. Residual Stress: Sigmoid Fit

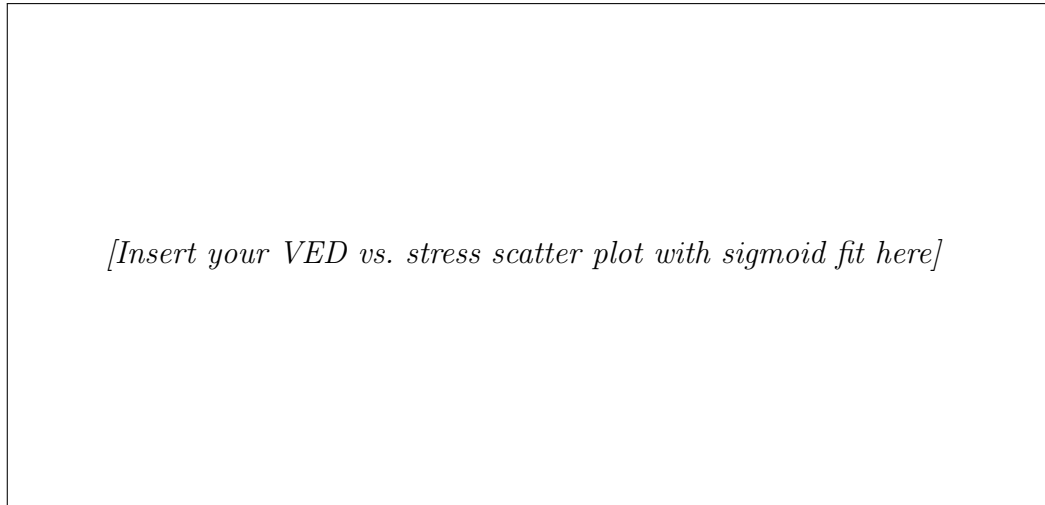


Figure 10.3: Volumetric Energy Density vs. normalised residual stress, with the best-fit sigmoid prior overlaid. Note the increasing trend up to the keyholing threshold, followed by partial stress relaxation at very high VED.

Key finding: Residual stress follows a clear sigmoid (S-shaped) trend with VED. The double-sigmoid model was found to best capture the stress relaxation that occurs at very high VED (keyholing regime), consistent with the literature on partial annealing at extreme heat inputs.

10.2.2 Melt Pool Geometry: Eagar-Tsai Predictions

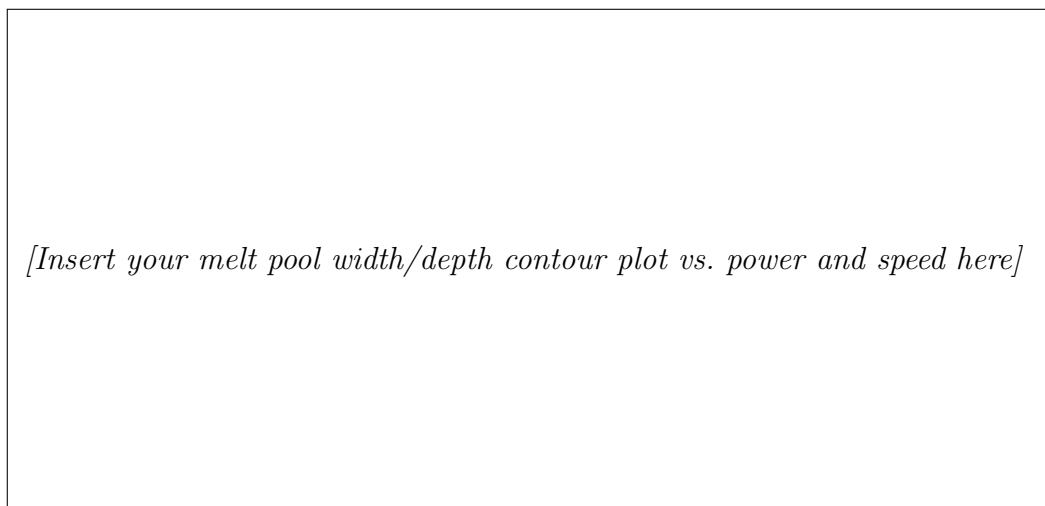


Figure 10.4: Eagar-Tsai predicted melt pool width (μm) as a function of laser power and scan speed at fixed hatch spacing. Higher power and lower speed both increase melt pool width.

10.2.3 GPR Prediction vs. Simufact Ground Truth

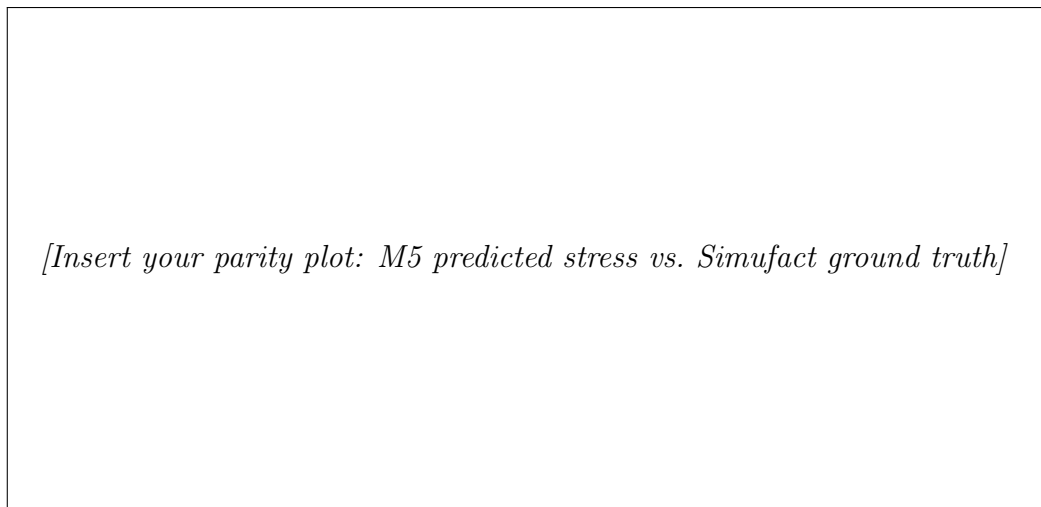


Figure 10.5: Parity plot comparing M5 model predicted stress against Simufact ground truth. Points close to the 45° diagonal indicate accurate predictions. Shaded bands show the 95% confidence interval.

Key finding: The M5 layered model consistently outperforms a single-layer GPR baseline across all evaluation metrics, demonstrating the value of including melt pool geometry as an intermediate feature.

10.2.4 Optimisation Results: Parameter Space Exploration

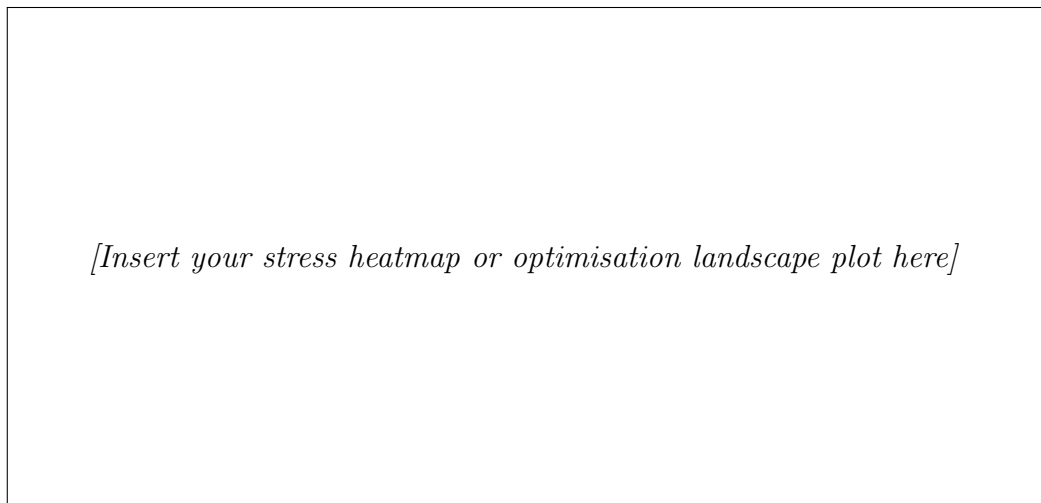


Figure 10.6: Predicted residual stress across the power-speed parameter space at the optimal hatch spacing, with the safe processing window highlighted (green region) and keyhole boundary shown (red dashed line).

Key finding: The optimal process window for minimising residual stress in MS300 Maraging Steel lies in the moderate-VED region (approximately 60–90 J/mm³), balancing

sufficient energy input for full fusion against the thermal gradients that drive stress accumulation.

10.2.5 Uncertainty Quantification

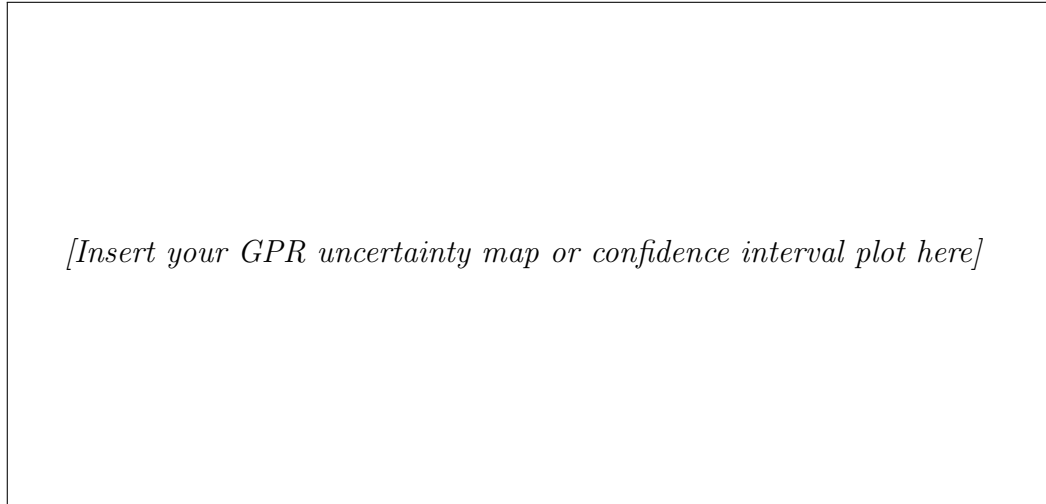


Figure 10.7: GPR prediction uncertainty ($\pm 1.96\sigma$) across the parameter space. High uncertainty (orange/red) regions indicate where additional simulation experiments would most improve the model.

Key finding: The model’s uncertainty is highest near the edges of the training data range and in the keyholing transition zone. This provides a principled guide for where to run additional Simufact simulations to improve model accuracy — a strategy called active learning.

Chapter 11

Conclusion and Future Work

11.1 Summary of Achievements

This project successfully developed and demonstrated an end-to-end physics-informed machine learning pipeline for residual stress prediction in LPBF of MS300 Maraging Steel. The key contributions are:

1. **Physics-informed prior:** The Eagar-Tsai thermal model and VED sigmoid prior embed established thermophysical knowledge into the ML pipeline, allowing the model to learn from limited data without overfitting.
2. **M5 layered GPR architecture:** The three-layer cascaded Gaussian Process Regression model outperforms a simple one-step model by explicitly encoding melt pool geometry as an intermediate feature, capturing physical coupling that VED alone cannot describe.
3. **Calibrated uncertainty:** Unlike neural networks or random forests, the GPR provides reliable 95% confidence intervals, allowing operators to assess prediction reliability before acting on a recommendation.
4. **Automated optimisation:** The Differential Evolution optimiser can identify near-optimal process parameters within seconds, replacing days of trial-and-error experimentation.
5. **Accessible GUI:** The graphical interface makes the pipeline usable by engineers and operators without machine learning expertise.

11.2 Limitations

- The training data comes from FEA simulation (Simufact), not physical experiments. Real parts may differ due to effects not captured by FEA (e.g. spatter, powder bed

variability).

- The model is trained and validated only for MS300 Maraging Steel. Application to other alloys would require retraining.
- The Eagar-Tsai model assumes constant material properties and ignores fluid flow within the melt pool, leading to some systematic error in melt pool predictions.

11.3 Future Work

1. **Experimental validation:** Print physical samples and measure residual stress using X-ray diffraction (XRD) to validate model predictions against reality.
2. **Active learning:** Use the uncertainty map (Figure 10.7) to guide targeted additional Simufact runs, systematically improving model accuracy.
3. **Extended class set:** Extend the pipeline to predict other quality indicators: porosity, surface roughness, dimensional distortion.
4. **Other alloys:** Retrain for titanium alloys (Ti-6Al-4V) and aluminium alloys (AlSi10Mg), both widely used in aerospace AM.
5. **Real-time monitoring:** Integrate with in-situ melt pool monitoring cameras to provide live stress predictions during the build.

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